

## WHAT A CALIPER IS AND HOW TO USE IT – 10 min

### Objective of Experiment No. 1

This experiment will help students become familiar with the caliper by performing various measurements on an object.

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### Equipment

- 1 × Vernier caliper (plastic, 0.05 mm precision)
  - 1 × Wooden parallelepiped
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### Description of the experiment

**Note:** For convenience, this experiment uses a single object from the kit, but it can be repeated with other, possibly more complex solids.

- To measure the width of the solid, place one of its bases (both have the same size) between the fixed jaw and the movable jaw of the caliper (see STEP 1). First, observe the main scale: in this case, the value is 4.2, which provides the main part of the measurement. Then observe the vernier scale (in this case, the value is 2), which provides the decimal part.

The total measurement is therefore:

$$42 \text{ mm} + 0.2 \text{ mm} = 42.2 \text{ mm} = 4.22 \text{ cm}$$

- To measure the height of the solid, proceed in the same way: place the longest side between the two jaws of the caliper (see STEP 2). Read the value on the main scale (in this case 10.2) and on the vernier scale (in this case 7).

The total measurement is therefore:

$$102 \text{ mm} + 0.7 \text{ mm} = 102.7 \text{ mm} = 10.27 \text{ cm}$$

- To measure the depth of the object, a cavity is required. In this case, one of the holes in the solid is used. Perform the measurement using the depth rod (see STEP 3). The reading is carried out in the same way as before. The measured values are:

$$18 \text{ mm} + 0.9 \text{ mm} = 18.9 \text{ mm} = 1.89 \text{ cm}$$

**Reminder:** To correctly read the vernier scale, select the mark that aligns with a mark on the main scale.

### Final observations

With this simple experiment, we used the caliper as a measuring instrument and observed the level of precision it can provide. To fully benefit from its usefulness and accuracy, it would be

advisable to perform measurements on more complex and smaller elements (e.g., bolts and fasteners).

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### Questions for the teacher

1. Consider a parallelepiped with a cylindrical hole that must be filled with a certain liquid. Is it possible to use the caliper as a tool to precisely calculate the amount of liquid required for this purpose?

### Cylinder volume formula:

$$V = \pi r^2 h$$

where:

- $r$  = radius of the cylinder base
  - $h$  = height of the cylinder
2. To practice further, measure the main dimensions of at least 5 objects in the laboratory. A table can be used to record the collected measurements.